



Flight Test Validation of Tandem Propeller Performance with Vertical Offset

Michael Foster¹ and Jessica DeMoor²
University of Dayton, Dayton, Ohio, 45469, United States

Sidaard Gunasekaran³
University of Dayton, Dayton, Ohio, 45469, United States

Jielong Cai⁴
Wichita State University, Wichita, Kansas, 67260, United States

Strategic placement of the rotors in a non-planar configuration in a multirotor vehicle can improve power efficiency. We extend our wind tunnel study on the investigation of the negatively staggered tandem propeller performance, aiming to validate the results through flight tests. The previous study conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) showed a power-saving benefit for the negatively staggered propeller configuration. An in-house-built flight test platform with various vertical propeller displacements was designed, built, and flown successfully in the current study. Flight test data confirmed up to 18% power saving for the negatively staggered propeller configuration when compared to the traditional non-staggered configuration, indicating the potential of the negatively staggered propeller configuration for practical applications. Furthermore, flight test data indicated that, at vertical offsets above $0.2d_z$, any increased power saving is minimal.

Nomenclature

D	=	Propeller Diameter, (m)	P_s	=	Single Rotor Power Required, (W)
d_x	=	$\frac{\text{Lateral Rotor Hub Offset (m)}}{D}$	P_t	=	Tandem Rotor Power Required, (W)
d_y	=	$\frac{\text{Longitudinal Rotor Hub Offset (m)}}{D}$	$P_{t/s}$	=	Power Ratio; $P_{t/s} = \frac{P_t}{P_s}$
d_z	=	$\frac{\text{Vertical Rotor Hub Offset (m)}}{D}$	R	=	Propeller Radius, (m)
J	=	Advance Ratio; $J = \frac{V_\infty}{nD}$	V_∞	=	Freestream Velocity, (m/s)
n	=	Propeller Rotational Speed, (RPS)	α	=	Angle of Attack, ($^\circ$)
P	=	Power, (W)	ΔP	=	Change in P ; $\Delta P = \frac{P_{\text{baseline}} - P_{\text{actual}}}{P_{\text{baseline}}}$

I. Introduction

Fixed-pitch propeller-driven multirotor aircraft are a growing commercial endeavor within the aerospace industry. As of 2024, nearly 800,000 multirotor unmanned aerial vehicles (UAVs) are registered with the Federal Aviation Administration (FAA) [1]. The shift from helicopters to multirotor aircraft for vertical takeoff and landing (VTOL) applications is driven by advances in electric motor, battery, and flight-control technologies. A common method to

¹ Undergraduate Student, Department of Mechanical and Aerospace Engineering, AIAA Member. michaelrwfoster@gmail.com

² Undergraduate Student, Department of Mechanical and Aerospace Engineering, AIAA Member. demoorj1@udayton.edu

³ Associate Professor, Mechanical and Aerospace Engineering Department, AIAA Senior Member. gunasekarans1@udayton.edu

⁴ Assistant Professor, Department of Aerospace Engineering, AIAA Member. jielong.cai@wichita.edu

increase the payload of eVTOL vehicles is to add more propellers, which also helps achieve lower disk loading, thereby improving efficiency and range [2]. These eVTOL air vehicles often operate in near-edgewise flight conditions, where the rotor discs are relatively parallel to the freestream. The downstream rotors of multirotor aircraft operate within the turbulent downwash of the upstream rotors. This interaction reduces efficiency due to factors such as wake interference, blade-vortex interactions, and flow unsteadiness. Consequently, the proximity between rotors significantly influences the magnitude of these effects.

In hover, overlapping or closely spaced rotors experience wake impingement, which alters the local inflow conditions and reduces thrust efficiency, as observed by Dingeldein [3] in wind tunnel studies on multirotor configurations. Similarly, Sweet [5] and Huston [6] examined how overlapping tandem rotor configurations incur significant aerodynamic penalties, particularly under hovering conditions. In forward flight, rotor wake interactions become more complex, often resulting in uneven loading and pitching moments that destabilize the aircraft. Otsuka et al. [4] demonstrated that such flow interference can induce pitching moments in quadrotor UAVs, while Prudden [7] investigated the broader aerodynamic implications for multirotors in forward flight.

Experimental and numerical investigations have further quantified how rotor-body and rotor-rotor interactions impact overall vehicle performance. Recent work by Herz et al. [9], McCrink et al. [10], and Atte et al. [11] has explored the compounded effects of these interactions on flight dynamics and efficiency. For tandem propellers, vertically offset configurations have shown potential to mitigate some wake interference effects. Cai et al. [13–15] systematically investigated these configurations in a wind tunnel, revealing how vertical and horizontal offsets influence performance, particularly in scenarios involving staggered or overlapping propellers. A well-established method of reducing rotor interference is to increase the horizontal spacing between upstream and downstream rotors [7, 15]. However, this approach increases the overall footprint of the aircraft, which may be disadvantageous for certain applications.

To maximize efficiency while preserving the compact footprint of tandem propellers in forward flight, the previous experimental campaign at the University of Dayton investigated the efficiency impact of a pair of tandem rotors within a wind tunnel [13, 14]. The tandem rotor pair was tested in three configurations: negative-offset, where the downstream rotor was positioned above the plane of the upstream rotor (Figure 1a), traditional, where both rotors were placed in the same plane (Figure 1b), and positive-offset, where the downstream rotor was positioned below the plane of the upstream rotor (Figure 1c).

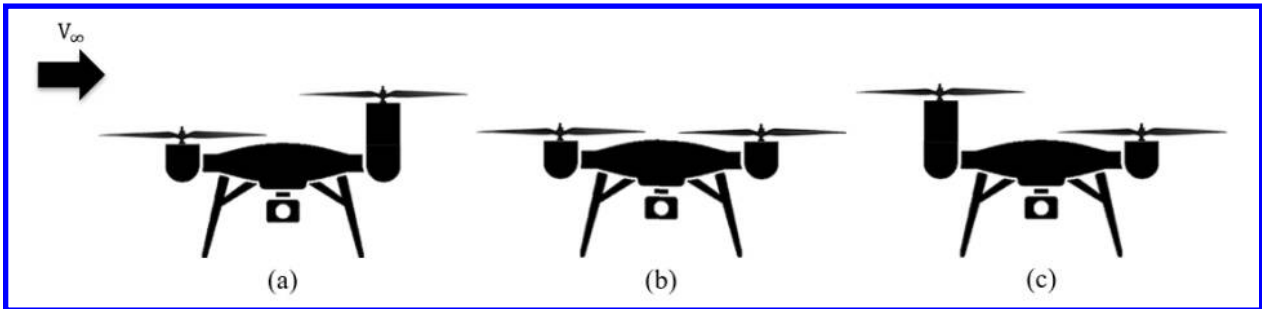


Figure 1. Schematic Showing (a) Negatively Staggered, (b) Traditional, and (c) Positively Staggered Rotor Orientations

Cai et al. [13] evaluated the performance of the negatively staggered configuration by comparing the power consumption of tandem rotor configurations (P_t) to that of two standalone rotors (P_s) producing equivalent thrust without any rotor interference. This result is illustrated in Figure 2 from [13]. In the figure, d_z represents the rotor diameter-normalized vertical offset between the rotors, and d_y is the rotor diameter-normalized horizontal distance between the rotors. A d_y of less than 1 indicates cases where the rotors overlap.

As shown in Figure 2, across all evaluated advance ratios, an increase in vertical offset corresponded to a decrease in the power consumption ratio, indicating reduced rotor interference. However, as the vertical offset increased, the positive impact on the power ratio diminished. When vertical offsets reached $d_z = 0.6$, the differences in power ratios became negligible. Although there was an inverse correlation between the power ratio and horizontal offset (d_y), the impact within each vertical offset group was minimal. Consequently, horizontal spacing of rotors can be minimized

by implementing vertical offsets to reduce rotor interference while maintaining the compact footprint of multirotor aircraft. However, all previous studies were conducted on a pair of tandem propellers in a wind tunnel. To extend the negatively staggered propeller configuration to practical applications, validation through flight tests on a multirotor aircraft is necessary. The current study utilized an in-house-built multirotor test vehicle to validate the experimental results from the previous study [13]. Additionally, flight tests were conducted to examine previously isolated factors, such as in-flight stability.

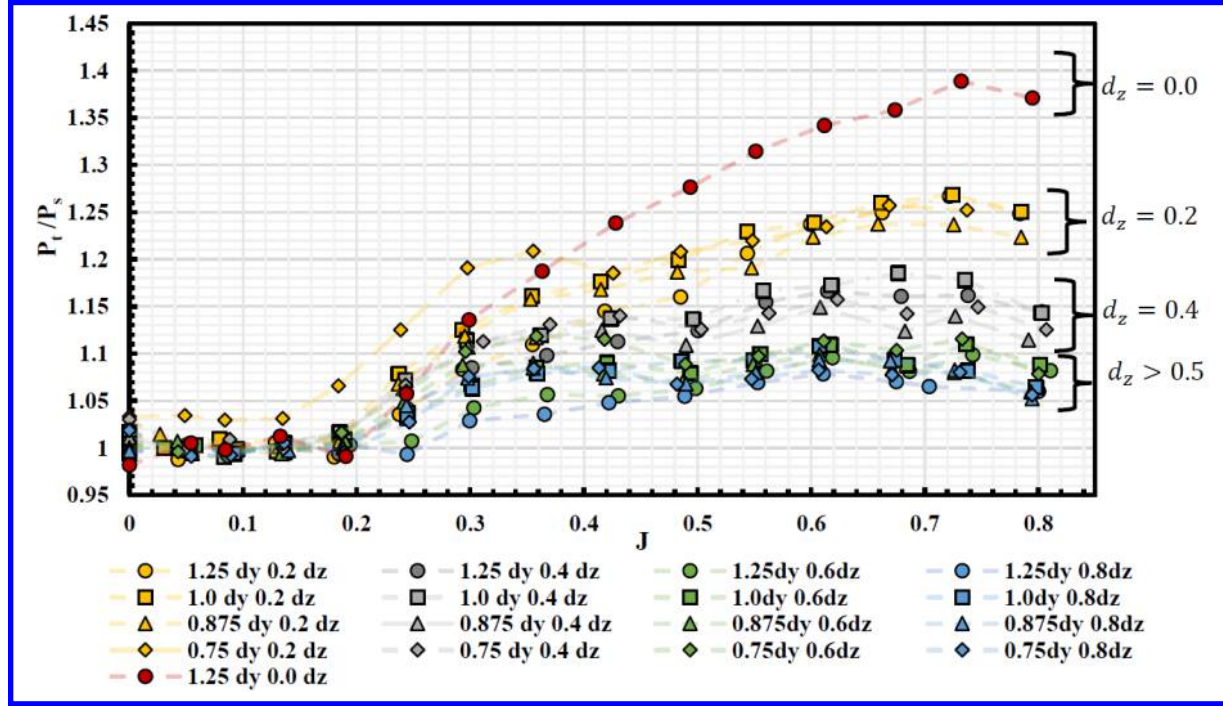


Figure 2. Ratio of P_t/P_s for all dz cases at $\theta = 90^\circ$ vs. J .

II. Experimental Setup

A. Flight Test Overview and Procedure

Flight tests were conducted at the City of Oakwood Old River Sports Complex at dawn under calm wind conditions to minimize statistical variance in collected data. A ground-based anemometer was positioned at the endpoint of the flight path to measure wind direction and speed throughout each run. An aerial image of the flight path used for testing, with a compass indicating the multirotor's flight path heading, is shown in Figure 3. The flight path consisted of at least three identical “down-and-back” segments, each comprising two runs at a fixed heading and speed, resulting in a total of six runs. Each run was 304.44 meters in length at an altitude of 15 meters above ground level (AGL). Depending on the direction, the multirotor had a heading of either 117° or 297° . At the endpoint of each run, the multirotor performed a triangular maneuver to rotate its heading by 180° , ensuring a yaw angle of 0° relative to the flight path at the start of the next run.

Flight testing was conducted to compare the power consumption of the multirotor with a vertical offset to a baseline configuration without vertical offset. The tests utilized GPS-controlled flight paths programmed in QGroundControl software [19] to maintain stable speed, heading, and altitude. This procedure ensured controlled flight paths with repeatable velocities and altitudes throughout testing.

Throughout testing, the mass was kept constant, both with and without varying amounts of vertical offset. Although the benefits of a vertical offset are realized only if they outweigh the drawbacks of the increased weight required to implement it, the flight testing had two primary objectives. The first was to validate the results of the previous study using flight test data [13], and the second was to determine whether stable edgewise flight could be achieved with a vertical offset. The corresponding test matrix for the initial testing is shown in Table 1.



Figure 3. Aerial View of Flight Path

Table 1. Preliminary Test Matrix for Multirotor Performance with Set Mass and Varying Vertical Offset

Propeller	Horizontal Offset (d_x)	Vertical Offset (d_z)	Flight Speed (m/s)	Runs per Configuration	Run Length (m)	Mass (g)
KDE – CF125	1.25	0, 0.2, 0.4, 0.6	3, 6, 9, 12	6	304.44	2455

As determined by the previous study [13], power consumption decreases as a function of longitudinal offset (d_y). For this reason, a normalized longitudinal rotor offset of 1.25 was selected, constrained by fuselage manufacturing limitations. A range of vertical offsets (d_z) between 0 and 0.6, in increments of 0.2, were chosen to compare the potential power savings of different configurations. Forward flight speeds of 3, 6, 9, and 12 m/s were selected to achieve varying velocities and vehicle incidence angles. For each configuration, three “down-and-back” paths were conducted to reduce statistical variation inherent in outdoor flight testing, resulting in six effective runs per case. Of these six runs, the three best runs were selected based on flight conditions, power consumption, and error minimization.

B. Multirotor Aircraft Test Platform

A custom multirotor aircraft was developed to incorporate the chosen longitudinal rotor offset and allow for rapid substitution of vertical offsets throughout experimental trials, as shown in Figure 4. The multirotor aircraft was powered by four KDE 2814XF-515 brushless motors paired with two-bladed 12.5” x 4.3 KDE propellers, both of which were utilized in the previous experimental study [13]. Four T-motor F-30A DShot electronic speed controllers (ESCs) were employed, with each ESC controlling an individual motor. To mitigate potential electrical noise passed to the ESCs, a 35V 1000 μ F capacitor was placed in parallel with the DC input of each ESC. Each motor and propeller assembly, along with its respective ESC, was attached to an off-the-shelf Lumenier QAV-R carbon fiber arm. All four arms were mounted at their roots to an in-house-built 3D-printed polycarbonate body frame. A Holybro PM07 14S Power Distribution Module was used to distribute power from the multirotor battery to all onboard electronics. The test vehicle's power source was an HRB 60C 6S 4000mAh 22.2V LiPo battery.

Flight control and data acquisition were handled by a Pixhawk 6C flight controller running PX4 [16], an open-source autopilot firmware with autonomous capabilities. The Pixhawk controlled speed, heading, and altitude of the vehicle to match the predetermined flight paths. During flight, data on attitude, speed, total power consumption, and

individual rotor rotational velocity were recorded at a rate of at least 4 Hz. The relatively low logging rate was due to the processing constraints of the Pixhawk flight controller. A forward view of the multirotor aircraft used in testing is shown in Figure 5.

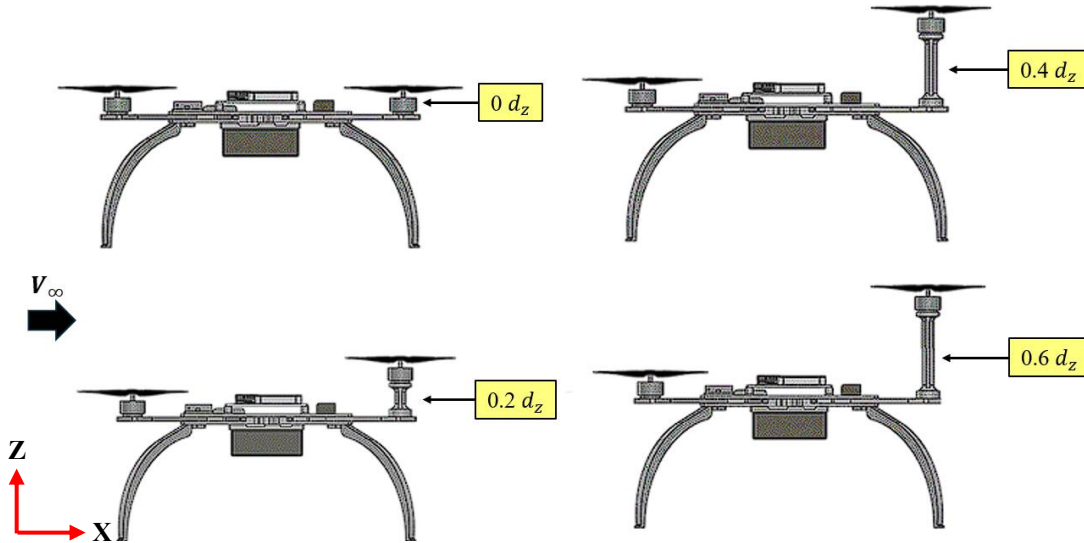


Figure 4. Side View of Test Platform with 0.0, 0.2, 0.4, and 0.6 d_z Configuration

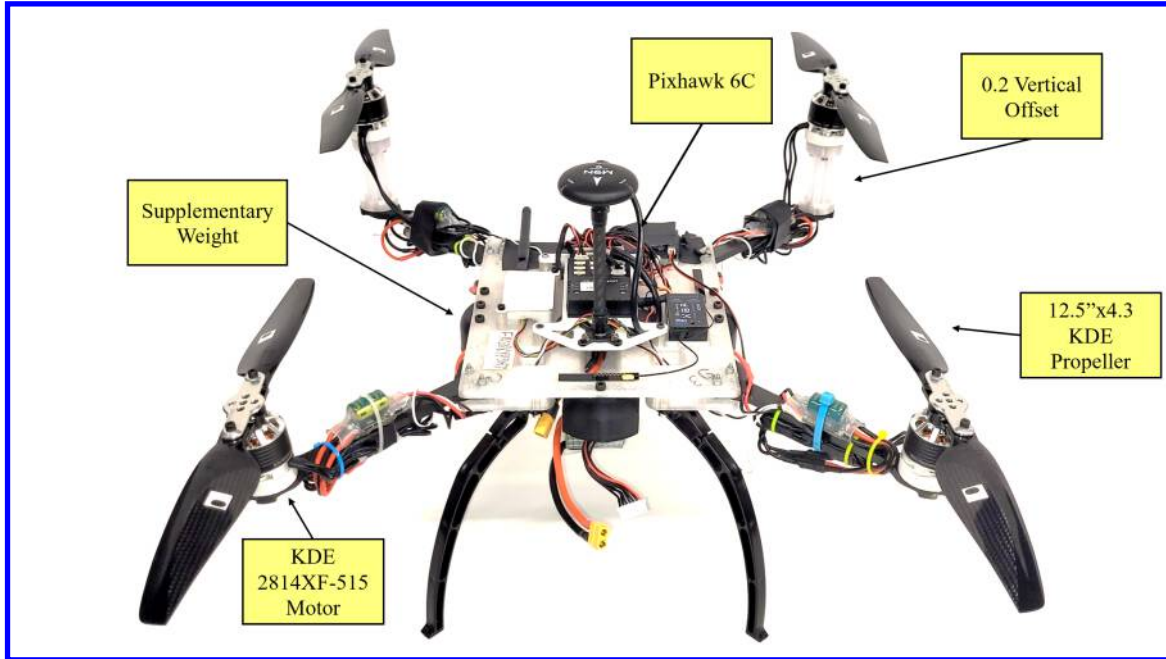


Figure 5. Front View of Multirotor Aircraft Test Platform with 0.2 d_z Vertical Offset

The addition of a vertical offset shifts the center of mass of the multirotor aircraft aft due to the added structure. To maintain identical rotational velocities for all rotors under hover conditions, the center of mass must align with the geometric center of the aircraft. To ensure this, an in-house-built balance-checking device was constructed. This device determines the alignment of the center of mass with the geometric center along the x- and y-axes. As shown in Figure 6, the aircraft's battery can be repositioned to counterbalance the additional weight introduced by the vertical offset.

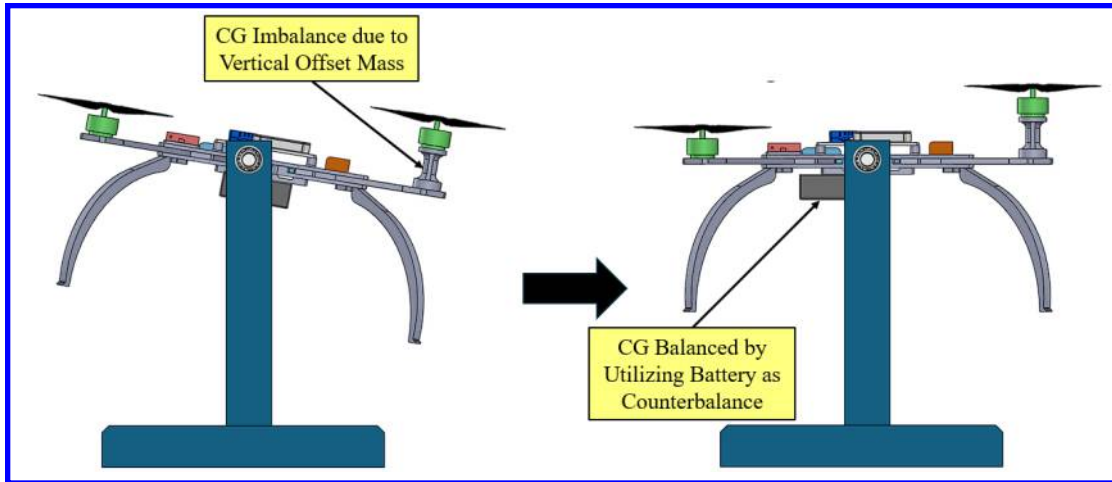


Figure 6. Aligning Center of Mass with the Locational Center Using the Balance-Checking Device

The addition of a vertical offset alters the mass of the multirotor. To accurately analyze power consumption across multiple stagger configurations, the mass of the multirotor was kept constant. The maximum takeoff weight (MTOW), corresponding to the configuration with the maximum vertical offset, was established as the baseline takeoff weight. For all other configurations, custom 3D-printed supplementary weights were used to increase the MTOW to this baseline value. These weights were designed to mount symmetrically to the front, rear, left, and right edges of the multirotor frame to maintain the center of mass along the x- and y-axes. The MTOW of the aircraft across all configurations was $2460 \text{ g} \pm 6 \text{ g}$, accounting for slight variations in battery weights, as multiple HRB 60C 6S 4000mAh 22.2V batteries were used during testing.

C. Anemometer

The anemometer used in the experimental study was a custom-designed, stationary setup featuring off-the-shelf wind speed (CALT YGC-FS-5V-V) and wind direction (CALT YGC-FX-5V-V) sensors. These sensors were mounted on a 3D-printed base designed in-house and secured on a tripod to collect data at a height of 1.88 meters AGL, as shown in Figure 7. During all test runs, ambient wind speed and direction were logged at a frequency of 1 Hz.

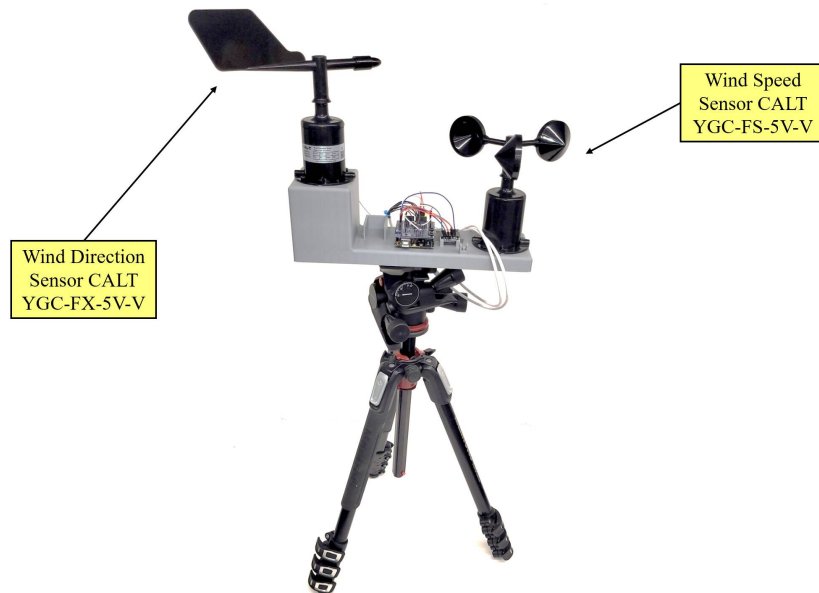


Figure 7. Side View of Anemometer

The sensors directly output voltage signals, which were converted into velocity magnitude and heading using

provided linear calibration equations. Each voltage reading corresponded to a specific wind speed or heading, ensuring precise wind behavior measurements. To verify the accuracy of the wind speed sensor, validation tests were conducted in the wind tunnel, as shown in Figure 8. Before flight tests, the anemometer was carefully aligned with true north during the calibration process to maintain directional accuracy.

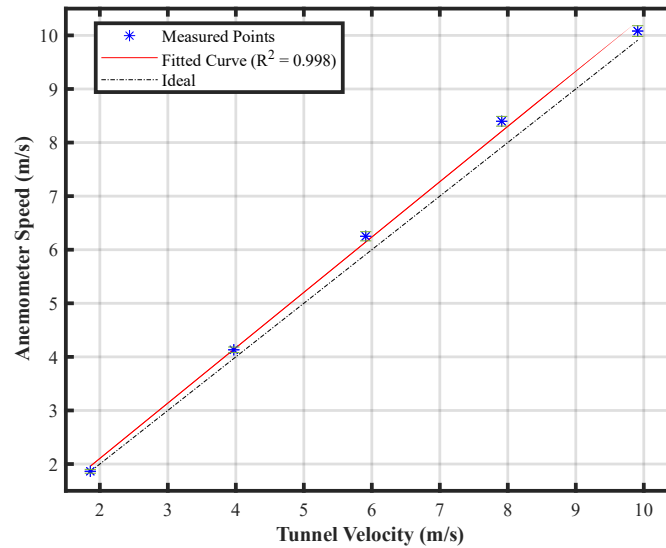


Figure 8. Difference between Tunnel Wind Speed and Anemometer Wind Speed

III. Results

All multirotor flight data used for analysis was recorded directly by the onboard Pixhawk flight controller. A representative flight path for a single configuration is shown in Figure 9. The flight path includes takeoff, aircraft acceleration, cruise, deceleration, and landing, as depicted in the figure. This study focuses on the power consumption of a multirotor during steady edgewise flight, characterized by steady-state velocity, altitude, and heading. Consequently, all other flight segments recorded by the flight controller were systematically excluded from analysis. To minimize human error, a custom MATLAB script was developed to identify and isolate flight segments meeting the steady-state criteria. The script analyzed flight test logs for all six runs of each configuration. First, a speed threshold was applied to identify segments where the speed remained within ± 0.5 m/s of the target value.

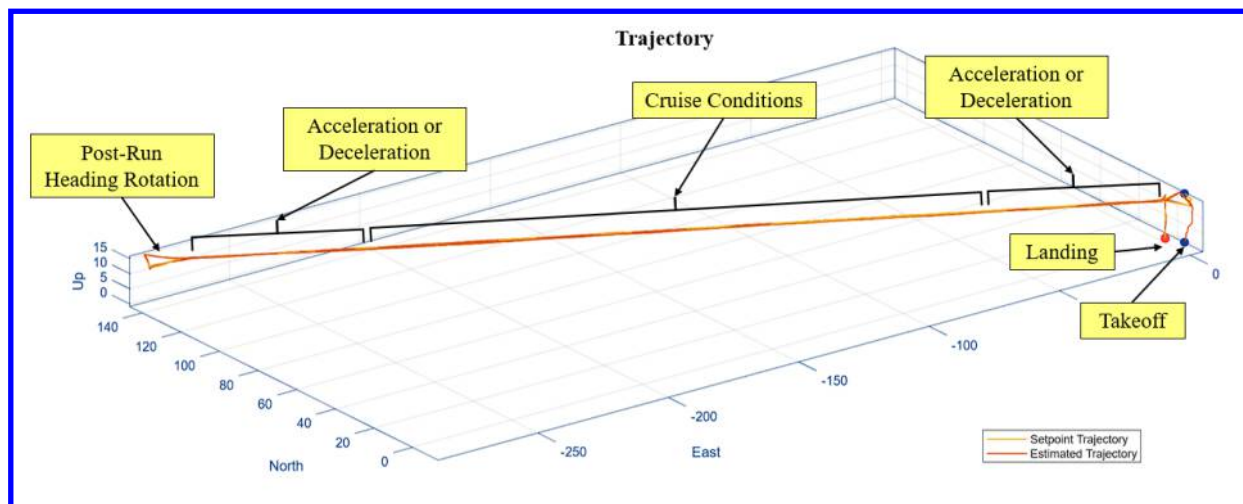


Figure 9. Representative Flight Path

A representative plot of flight speed data for a single run, with isolated cruise segments, is shown in Figure 10, where cruise conditions are marked by dashed lines. Each identified segment was evaluated to ensure altitude stability within a 2-meter range. Runs failing to meet this criterion were excluded from further analysis. The remaining runs for each configuration were then assessed to determine the three with the lowest average power consumption, which were selected for inclusion in the results presented in this paper.

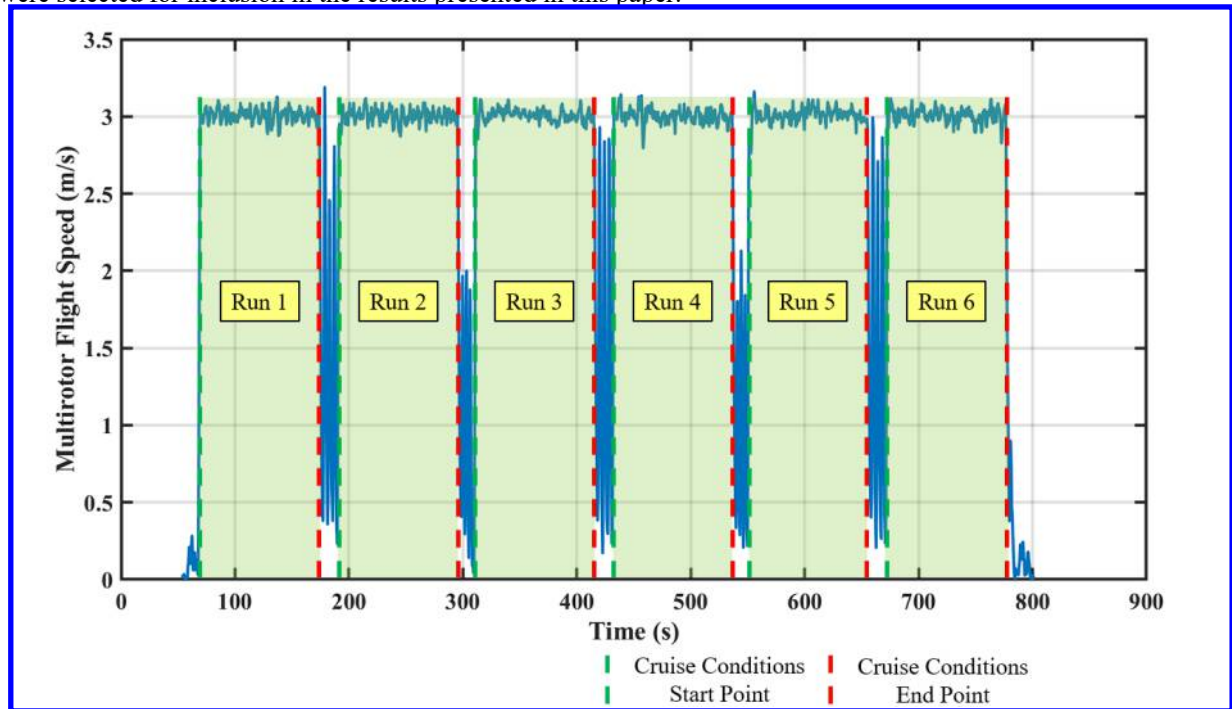


Figure 10. Representative Speed vs. Time Plot for Six Runs

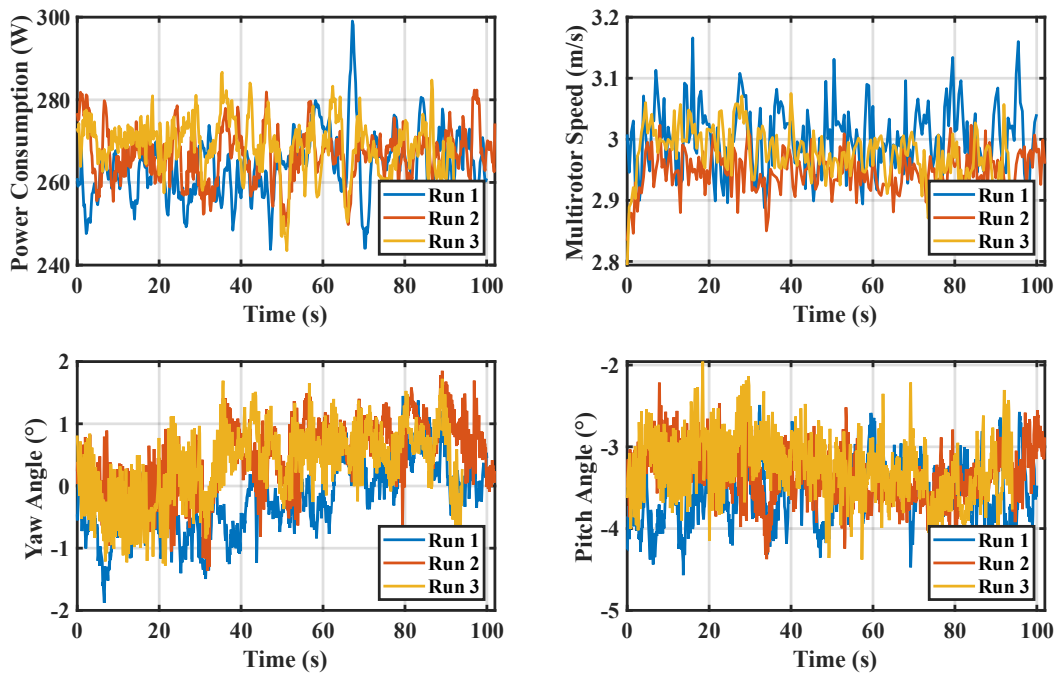


Figure 11. Representative Parsed Datasets for Best Three Runs of a Given Configuration at 3 m/s

Figure 11 provides a representation of the unfiltered speed, power, and attitude flight data for the cruise segments of the best three runs of a given configuration. As shown, the unfiltered datasets contain sporadic peaks that manifest as outliers during the statistical analysis. For flight tests conducted at a speed of 3 m/s, the average cruise duration was 99.25 seconds. As flight speed increased, longer acceleration times reduced cruise duration, with tests at 6 m/s averaging 40.85 seconds, tests at 9 m/s averaging 23.55 seconds, and tests at 12 m/s averaging 12.26 seconds.

A. Ambient Conditions

The time series in Figure 11 shows that the flight data align well across the three runs, indicating a high degree of consistency in multirotor behavior under the test conditions. However, closer inspection reveals minor fluctuations and small peaks and dips in the traces, particularly in power consumption and velocity. These fluctuations could be attributed to subtle changes in ambient wind conditions during the tests. To fully assess these variations, the quality of the three selected runs for each configuration and the observed ambient conditions were validated using anemometer data. Figure 12 presents the average ground-based wind speeds for all test runs. As shown, average wind speeds were at or below 1 m/s, with maximum wind gusts never exceeding 3.5 m/s at any time, indicating a calm and stable test environment.

A box plot analysis highlights differences in ambient wind speed between the test cases. For the 3 m/s, 6 m/s, and 9 m/s cases, the notches representing the 95% confidence intervals around the median wind speeds are narrow, suggesting a high degree of confidence in the median values and reflecting consistently low and steady winds. In contrast, the 12 m/s case shows a noticeably wider notch, indicating greater variability in wind speed during that test. This variability may be due to the 12 m/s tests being conducted entirely on a single day, whereas the other cases spanned multiple days, allowing for more fluctuations in wind patterns. Furthermore, the wind speeds show a positive skew at 3 m/s and 6 m/s, with the median closer to the lower quartile and outliers extending towards higher wind speeds. At 9 m/s and 12 m/s, the range of data becomes more symmetric, though slight variability remains. These patterns indicate variability in ambient wind conditions during testing, which may influence multirotor flight stability and performance across tested configurations and flight speeds.

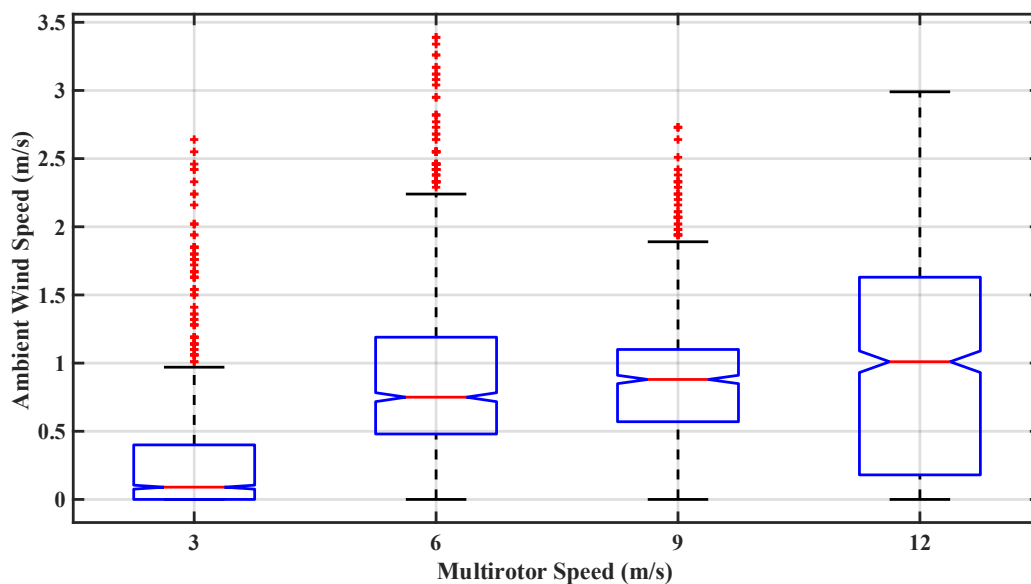


Figure 12. Ambient Wind Speed of all Test Runs

For the 3 m/s, 6 m/s, and 9 m/s cases, the interquartile range (IQR) is relatively small, confirming the tight clustering of the middle 50% of the wind speed data. This reinforces the consistent nature of the test environment for these conditions. In contrast, the 12 m/s case shows a larger IQR, indicating greater spread and variability in wind speeds encountered during the tests. This variability may partly explain the slightly more noticeable fluctuations in

vehicle performance data for this case. Outliers are more prominent in the 3 m/s, 6 m/s, and 9 m/s cases. These outliers, represented by individual data points beyond the whiskers of the boxes, likely correspond to brief gusts of wind that exceeded the average conditions during those test runs. Conversely, the 12 m/s case does not display any outliers. As mentioned earlier, tests at this flight speed were conducted on a single day with more consistently-low wind speeds. These observations provide further context for the minor fluctuations and peaks observed in Figure 11. The occasional gusts of wind captured by the outliers in the wind speed data are likely responsible for these variations in vehicle performance.

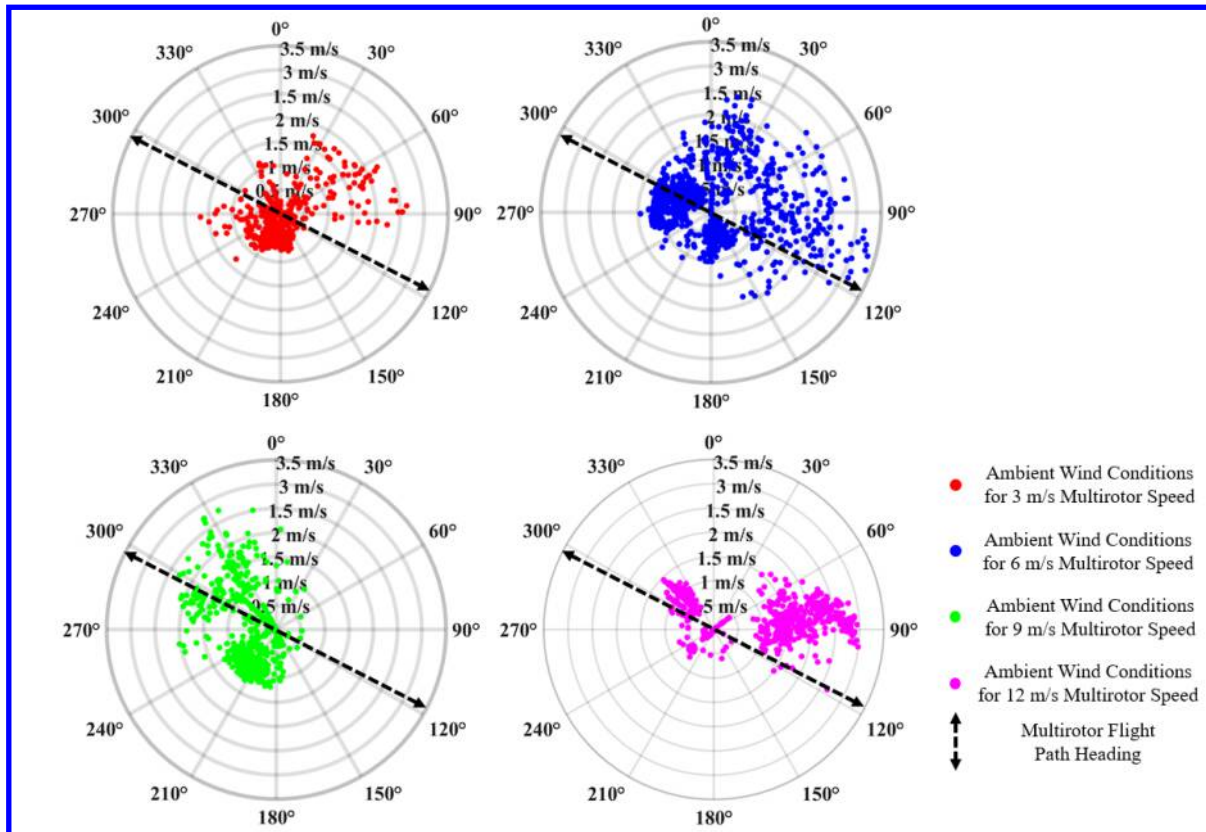


Figure 13. Ambient Wind Conditions and Flight Path for all Test Cases

Figure 13 illustrates the instantaneous wind speed and direction for all selected test runs, with the multirotor's flight path heading overlaid. As shown, the wind direction generally aligns with the flight path, acting either as a headwind or tailwind depending on the specific test run's flight direction. Among all multirotor speeds, the 6 m/s test runs exhibit the largest variation in wind direction and the greatest deviation from the flight path heading. The wide spread of data points indicates gusty conditions, although crosswinds remain calm, with gusts never exceeding 3.5 m/s. In contrast, the 12 m/s test runs show two distinct groupings of wind direction, one from the East and one from the West, each with a small spread. These groupings suggest gusts with 180° heading changes, which could momentarily push the multirotor off its flight path as it reacts to these changes. Despite these variations, wind conditions remained calm across all test runs. This indicates that statistical variations due to ambient conditions were minimal, enabling consistent and repeatable data for attitude, flight speed, and altitude.

B. Multirotor Vehicle Data

Figure 14 depicts the statistical variation in multirotor flight speeds across all test cases. As shown, the multirotor maintained a consistent speed with minimal deviation from the setpoint at all tested flight speeds. The median speed closely aligns with the target speed in each case, indicating that the vehicle achieved the desired cruise velocity with a high degree of accuracy. The narrow notches around the medians suggest a high level of confidence in these measures

of central tendency, reflecting consistent and repeatable speed control. However, differences in the overall spread of the data are observed across the test cases. The 3 m/s and 9 m/s conditions exhibit the tightest IQRs, indicating that the middle 50% of the speed data is closely clustered around the median. This suggests the vehicle demonstrated the most stable and consistent speed performance under these conditions. In contrast, the 6 m/s and 12 m/s cases show slightly larger IQRs, reflecting comparatively higher variability in flight speeds, likely due to increased environmental influences, such as higher average wind speeds and gusts.

The 3 m/s test cases exhibit several outliers, with data points falling well outside the overall distribution. These outliers likely correspond to brief speed increases caused by sudden wind disturbances, despite the relatively low average wind speeds for this case. In contrast, the 12 m/s test cases have the fewest outliers, even though they encountered the highest average wind speeds. This suggests that the vehicle was more effective at counteracting wind effects at the highest cruise speed, maintaining a more consistent velocity profile.

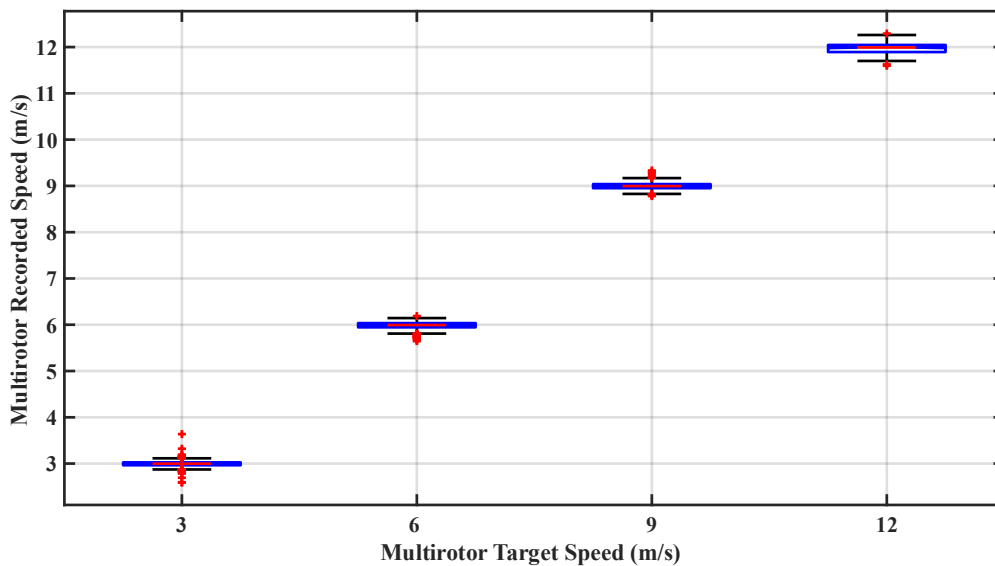


Figure 14. Flight Speed Distribution for all Test Cases

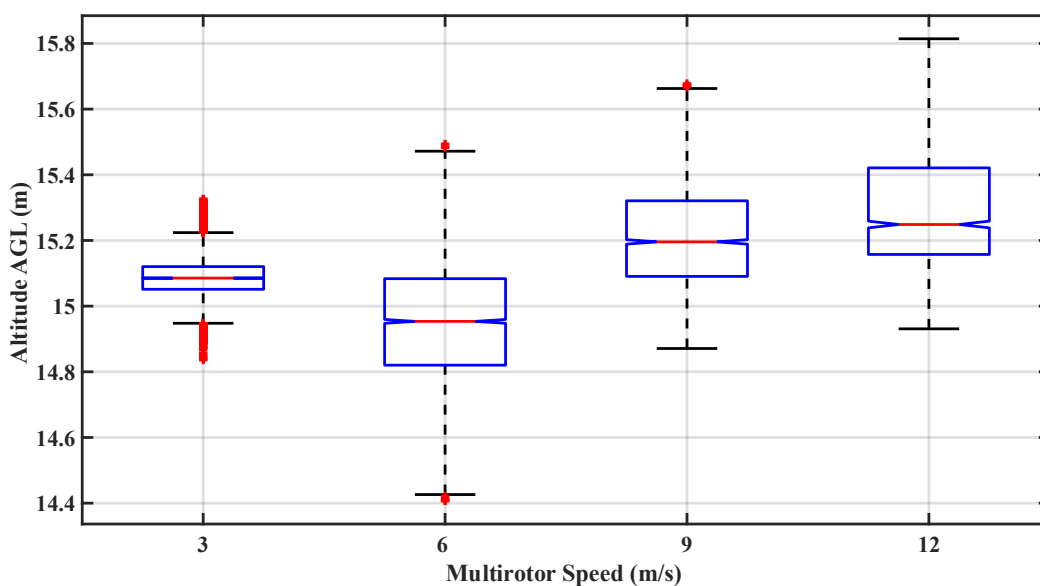


Figure 15. Flight Altitude of all Test Runs for all Flight Speeds

Figure 15 illustrates the statistical variation in altitude data for the multirotor across all test cases. As shown, the multirotor maintained a stable mean and median altitude, ranging between 14.95 and 15.25 meters AGL, depending on the tested flight speed. For each speed, the altitude variation from the average never exceeded 0.6 meters, approximately $\pm 4\%$. The 6 m/s test cases exhibited the largest IQR, indicating greater altitude variability during those tests. This observation aligns with the previously noted increase in speed variability at 6 m/s, suggesting a likely connection between the two factors. Conversely, the 3 m/s test cases had the smallest IQR, indicating the tightest clustering of the middle 50% of the altitude data. This may result from the vehicle's higher momentum at increased speeds, making it more resistant to altitude disturbances. However, the narrow notches around the medians across all cases suggest a high level of confidence in these central tendency measures. Furthermore, 3 m/s test cases displayed a higher number of outliers, likely due to the consistent sampling rate used for data acquisition across all velocity cases. At lower cruise speeds, the number of data points collected was nearly four times greater than in the 12 m/s test cases. As such, the number of outliers in the 3 m/s data is more prominent than the other cases.

The uniformity of flight stability is further supported by the variability of the yaw and pitch angles of the multirotor. Figure 16 shows the variation in yaw during cruise conditions for each speed tested across all vertical offset cases. As shown, the mean and median yaw angles during cruise conditions were consistently near zero degrees, indicating the vehicle maintained a steady heading throughout the tests. Variations in instantaneous yaw during cruise conditions were not part of the flight path but rather represented the flight controller's responses to wind gusts.

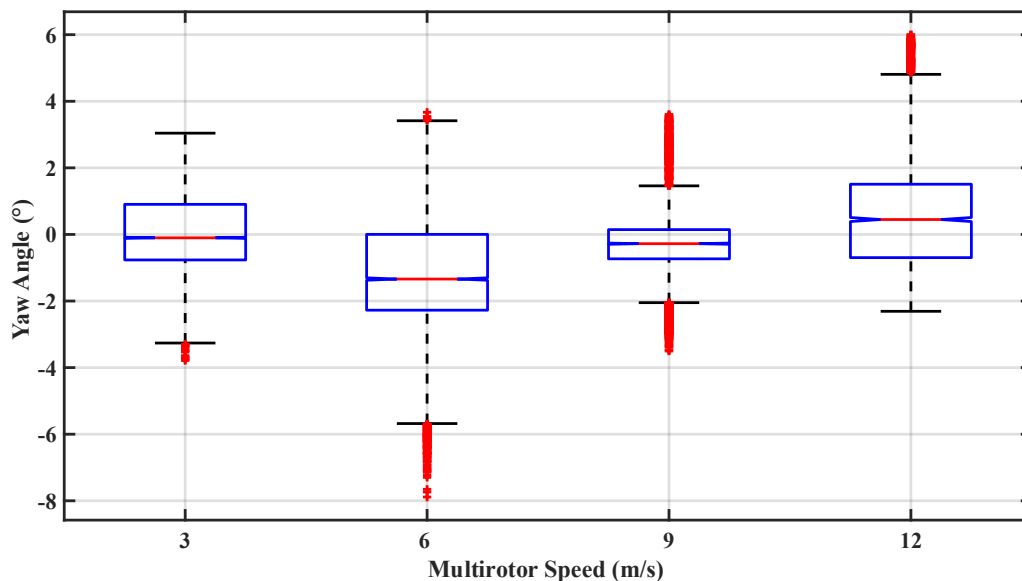


Figure 16. Yaw Angle of all Test Runs for all Flight Speeds

Figure 17 illustrates the distribution of pitch angle versus flight speed for all test runs across all configurations. A negative value indicates a pitch-down attitude of the multirotor. As expected, regardless of the vertical offset configuration, the pitch angle became more negative as flight speed increased, reflecting the larger downward-pitch angles required to maintain stable cruise flight at higher speeds. At flight speeds of 3 m/s and 6 m/s, the pitch angle remained relatively constant across all vertical offsets, although the pitch angle at 6 m/s was lower than at 3 m/s. At higher velocities of 9 m/s and 12 m/s, the pitch angle became increasingly negative as a function of both vertical offset and cruise speed. This variation in pitch angle due to vertical offset may result from the shift of the vehicle's center of gravity along the +z-axis (CG in x and y axes remained the same for all cases) caused by the offsets used in the rear rotors. In Figure 17, at a flight speed of 3 m/s, the IQR boxes are relatively short, indicating that the multirotor's pitch angle exhibited low variation around the median, reflecting a stable pitch attitude. The narrow notches further indicate a high degree of confidence in the median values for this speed. However, at higher pitch angles associated with increased speeds, the length of the boxes grows, signifying greater variation in pitch angles at these conditions.

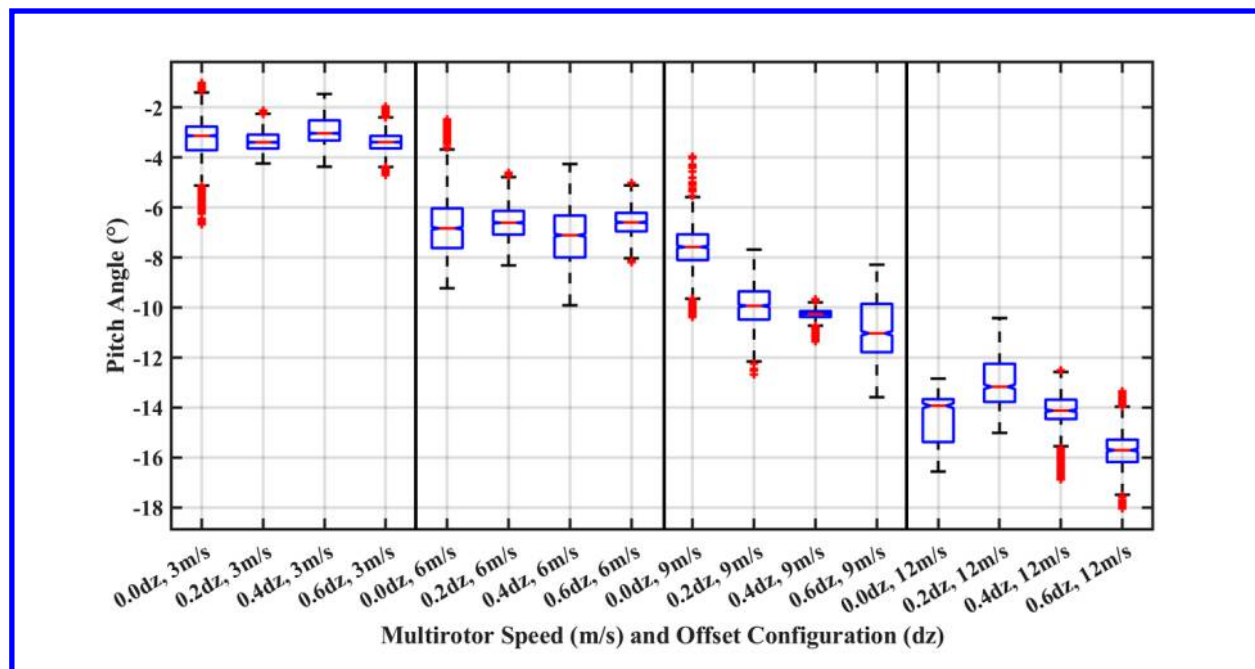


Figure 17. Pitch Angle of all Test Runs for all Flight Speeds

The consistency in multirotor attitude, flight speed, and altitude data across all test cases confirms that the aircraft maintained stable edgewise flight regardless of vertical offset. Additionally, this uniformity validates the comparison of power consumption across all cases. The power consumption for the three selected runs of each test case was averaged and plotted against flight speed, as shown in Figure 18.

Across all vertical offsets, power consumption decreases with increasing flight speed, indicating improved rotor efficiency, until a flight speed of 9 m/s, where efficiency begins to decline significantly. The results demonstrate that the power consumption benefits of a vertical offset are evident across all tested flight speeds and become more pronounced at higher speeds. The confidence in these benefits is supported by the relatively small error bars for each test case. However, for vertical offsets of 0 and 0.2 at 9 m/s, the error bars are slightly larger than average, likely due to ambient conditions during those specific runs.

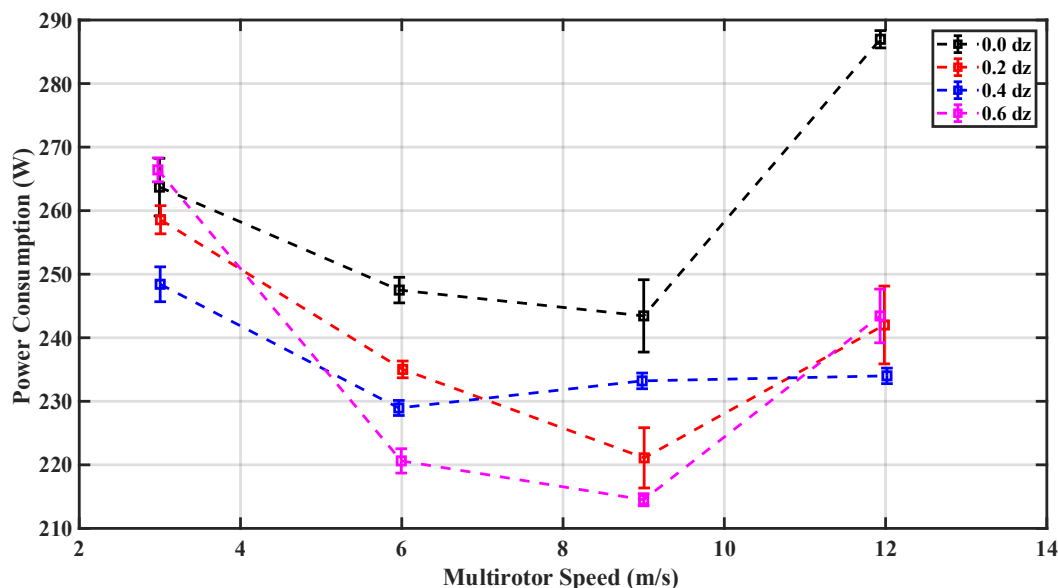


Figure 18. Power Consumption of all Test Runs for all Flight Speeds

At a flight speed of 3 m/s, the decrease in power consumption with increasing vertical offset is minimal. Additionally, at this same flight speed, a vertical offset of 0.6 was found to increase the power consumption of the multirotor compared to the configuration without any vertical offset. This increase in power consumption was consistently observed across multiple test runs conducted on different flight-test days, making it unlikely that the result was due to testing error. Instead, this outlier may be attributed to a possible harmonic interaction or increased drag caused by the vertical offset geometry, which outweighs the benefit of reduced rotor interaction at such a low cruise speed.

The average power consumption was normalized as a percent difference from the baseline configuration with a vertical offset of 0, as shown in Figure 19. Generally, the percent change in power consumption increased with velocity, indicating that while vertical offsets improve efficiency across all tested flight speeds, the benefits become more pronounced at higher speeds. This trend may be attributed to the increase in pitch angle correlated with higher flight speeds, where the rear rotors experience greater downwash from the front rotors in the absence of a vertical offset. Additionally, flight test data for a vertical offset of 0.2 shows the most uniform trendline of increased power savings compared to the 0.4 and 0.6 configurations, which exhibit more erratic trends. Across the vertical offset cases of 0.2, 0.4, and 0.6, the power savings remain relatively consistent with each other. This suggests that while introducing a vertical offset of 0.2 significantly improves power savings, further increasing the offset yields diminishing returns.

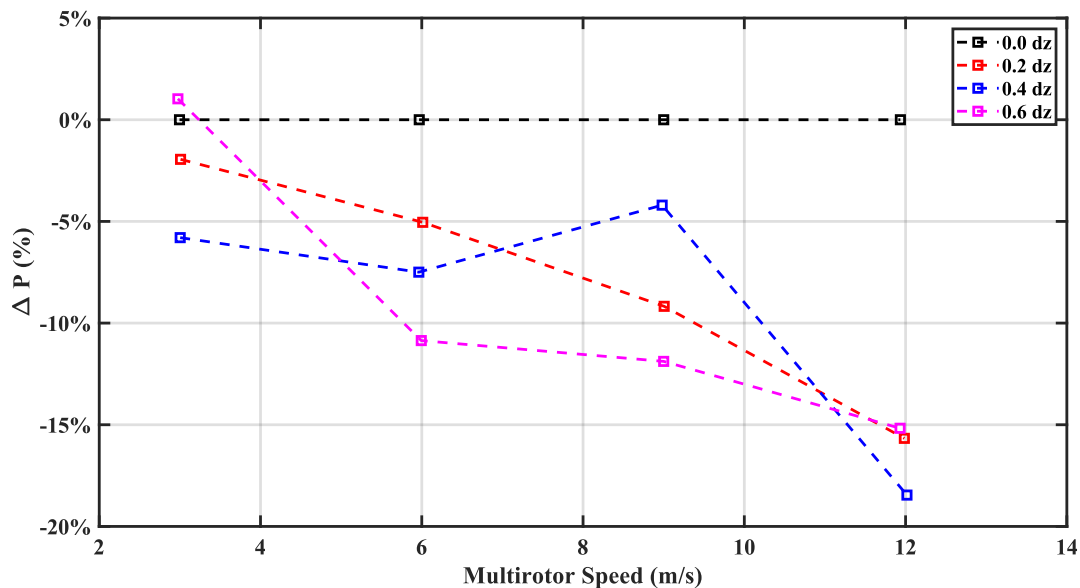


Figure 19. Change in Power Consumption with Varying Vertical Offset for all Flight Speeds

At a flight speed of 12 m/s, the maximum power savings compared to the baseline was 18%, achieved with a vertical offset of 0.4. At the same speed, vertical offset configurations of 0.2 and 0.6 both achieved power savings of approximately 15%. In the previous study, it was observed that power savings generally increased with vertical offset across flight speeds. However, in this case, the reduced power savings for the 0.6 offset compared to the 0.4 offset may be attributed to testing errors, increased drag from the additional components required to offset the rear rotors, or harmonic interactions, particularly given the flexibility of the multirotor airframe.

The power consumption data also shows a notable increase in power savings starting at a vertical offset of 0.2. This suggests that beyond a vertical offset of 0.2, further reductions in power consumption are minimal. The diminishing returns on power savings may result from increased drag due to vertical offset geometry or aeroelastic interactions leading to harmonics and higher data uncertainty.

The power savings achieved with a vertical offset are only beneficial if they outweigh the disadvantages of the increased takeoff weight caused by the offset. In a multirotor study that accounts for the weight impact of vertical

offsets, power savings may decrease or even reverse for offsets exceeding 0.2. In practical terms, a multirotor with a normalized vertical offset of 0.2 can achieve significant power savings with minimal added weight due to the offset geometry.

The advance ratio, J , is a useful metric to relate the forward speed of the multirotor to the average rotational velocity of the rotors (RPS). For the three selected test runs of each configuration and flight speed, the advance ratio of both rear rotors and both front rotors were averaged and plotted against flight speed, as shown in Figure 20. The advance ratios for the $0 d_z$ configuration at 3 m/s and for the $0.6 d_z$ configuration at 3 m/s and 6 m/s are not reported due to a loosening of the wires in the onboard circuitry during the test flights, likely caused by increased vibrations.

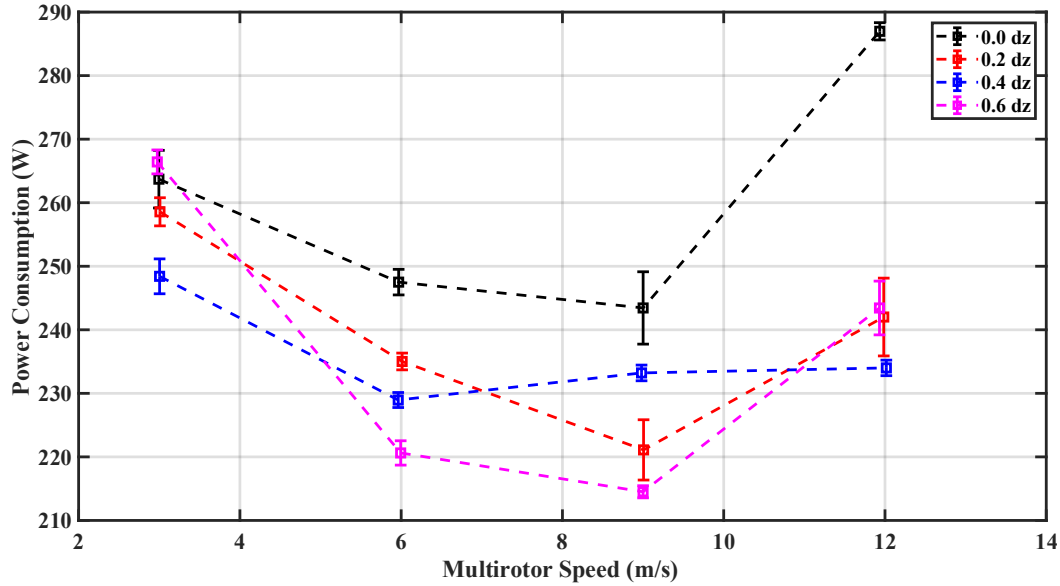


Figure 20. Advance Ratio as a Function of Multirotor Flight Speed

Evaluating the remaining data, the advance ratio increases with flight speed, corresponding to higher rotor velocities. Additionally, for all vertical offsets, the advance ratio of the front rotors is consistently higher than that of the rear rotors at all flight speeds. This trend arises because the rear rotors rotate faster than the front rotors to maintain forward flight. Despite the observed decrease in power consumption with increasing vertical offset, the rotational velocity of both the front and rear rotors remains similar across all vertical offset configurations at a given flight speed. This similarity highlights that even minimal changes in rotational velocity can have a significant impact on the overall power consumption of the multirotor.

To compare the flight test results with the results of the previous wind tunnel study [13], the percent change in power consumption relative to the baseline configuration with 0 vertical offset was plotted against the advance ratio in Figure 21. Since rotor power consumption was not recorded independently, the average advance ratios of the front and rear rotors were combined to obtain a single average advance ratio for the multirotor for each configuration and flight speed. Advance ratio data was not recorded for the $0 d_z$ configuration at 3 m/s and the $0.6 d_z$ configuration at 3 m/s and 6 m/s due to inconsistent logging of rotor rotational velocities. As shown in Figure 2, the wind tunnel tests indicated that at a horizontal offset of $1.25 d_y$, vertical offsets begin to improve power consumption at advance ratios greater than 0.18. Consistent with this, the included data shows power savings of approximately 5% at low advance ratios. As the advance ratio increases, corresponding to higher flight speeds, power consumption decreases further to 18%, affirming the wind tunnel results from the previous study.

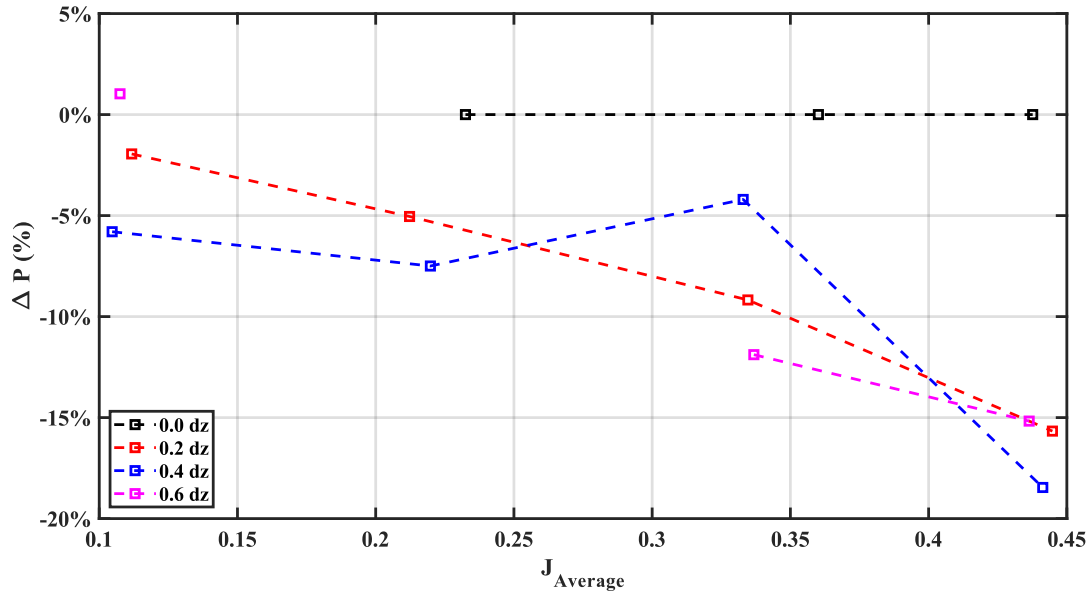


Figure 21. Change in Power Consumption as a Function of Advance Ratio

IV. Conclusion

Flight testing of the multirotor aircraft test platform demonstrated significant performance improvements with vertical offset in a negatively staggered configuration compared to the baseline configuration without stagger. Up to an 18% reduction in power consumption was observed during flight tests at speeds up to 12 m/s. The improvement, achieved by implementing a vertical offset in a negative stagger configuration, increases flight efficiency by reducing the rear rotors' exposure to downwash from the front rotors. This reduction, attributed to decreased rotor interference, validates the wind tunnel results from the previous study [13]. Furthermore, flight test data revealed a significant decrease in power consumption with a vertical offset of 0.2, while higher offsets produced diminishing returns. The weight penalty due to a small offset is modest for an upwards of 15% performance gain. The aircraft maintained stable flight with normalized vertical offsets ranging from 0.2 to 0.6 in both autonomous and manual modes, confirming the platform's reliability for practical applications. By incorporating a vertical offset, the platform achieves efficiency gains without significantly increasing its footprint, an important factor for operations in confined environments. These findings extend the feasibility of negatively staggered configurations to real-world applications, offering a balance between efficiency and compact design. Future work will focus on optimizing the ideal combination of horizontal and vertical offsets to further improve performance and flight endurance.

V. References

- [1] "Drones by the Numbers," FAA. [Online]. Retrieved from faa.gov/uas [Accessed: November 2024].
- [2] Leishman, J. G., "Principles of Helicopter Aerodynamics", Cambridge University Press, 2006.
- [3] Dingeldein, R. C., "Wind Tunnel Studies of the Performance of Multirotor Configurations," NACA TN 3236, August 1954
- [4] Otsuka, H., and Nagatani, K., "Pitching moment caused by rotor flow interference on a small quad-rotor unmanned aerial vehicle in a uniform flow," in Transactions of the Japan Society for Aeronautical and Space Sciences, 2018, vol. 61, no. 2, pp. 87–96.
- [5] Sweet, G. E., "Hovering measurements for twin-rotor configurations with and without overlap," Natl. Aeronaut.

Sp. Adm., 1960.

[6] Huston, J., “Wind-Tunnel Measurements of Performance, Blade Motions, and Blade Air Loads for Tandem-Rotor Configurations with and without Overlap,” Hampton, Va, 1963.

[7] Prudden, S. L., “Rotor Aerodynamic Interaction effect for multirotor unmanned aircraft system in forward flight” Ph.D. Thesis, RMIT University, Australia, September 2020

[8] Tsaltas, J. D., “Trim Solutions of Multirotor Vehicles using a Fast Performance Prediction Method,” Master’s Thesis, Ryerson University, December 2018.

[9] Atte, A., Wylie, D., and Rauleder, J., “Experimental Evaluation of Multi-Rotor Aerodynamic Interactions”, VFS 78th Annual Forum, May 2022.

[10] McCrink, M., Seth, D., and Herz, S., “Flight Test Measurement of Quadrotor Performance at Varying Sideslip Angles” VFS 78th Annual Forum, May 2022.

[11] Herz, S, Atte, A, Seth, D, Rauleder, J, and McCrink, M., “Impact of Rotor–Rotor and Rotor–Body Aerodynamic Interactions on Quadrotor Vehicle Performance”, VFS 77th Annual Forum, May 2023.

[12] Kostek, A. A., Braukmann, J. N., Loßle, F., Miesner, S., Visingardi, A., Boisard, R., Riziotis, V., Keßler, M., and Gardner, A. D., “Experimental Investigation of Quadrotor Aerodynamics with Computational Cross-Validation” VFS 77th Annual Forum, May 2023.

[13] Cai, J., Gunasekaran, S., and OL, M. V., “Vertically Offset Overlapping Propellers in Tandem Configuration” AIAA SciTech 2023, National Harbor, MD, January 2023, AIAA 2023-2464.

[14] Cai, J., Gunasekaran, S., and OL, M., V., “Investigation of Positively Staggered Vertically Offset Propellers in Tandem”, AIAA SciTech 2024, Orlando, FL, January 2024, AIAA 2024-1510

[15] Cai, J., “Changes in Propeller Performance Due to Rotor and Ceiling Proximity in Forward Flight” Ph.D. Dissertation, University of Dayton, May 2024.

[16] “Holybro Pixhawk 6C,” PX4 User Guide. [Online]. Available:

https://docs.px4.io/main/en/flight_controller/pixhawk6c.html [Accessed: May 2024].

[17] “QGroundControl User Guide,” QGroundControl. [Online]. Available:

<https://docs.qgroundcontrol.com/master/en/qgc-user-guide/index.html> [Accessed: May 2024].